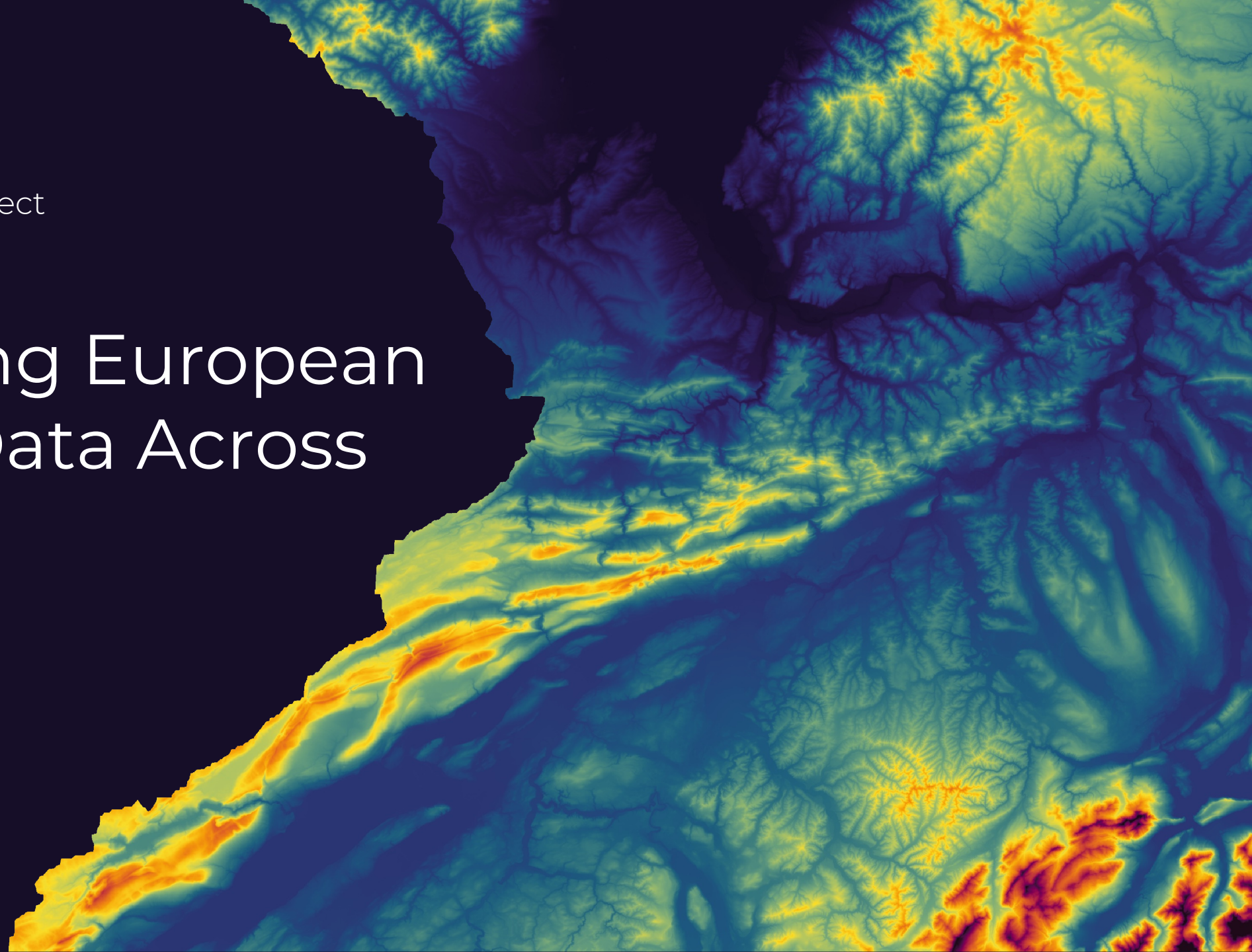
GEOIT1501: Synthesis Project
Geomatics Day 2026

Harmonising European Elevation Data Across Borders

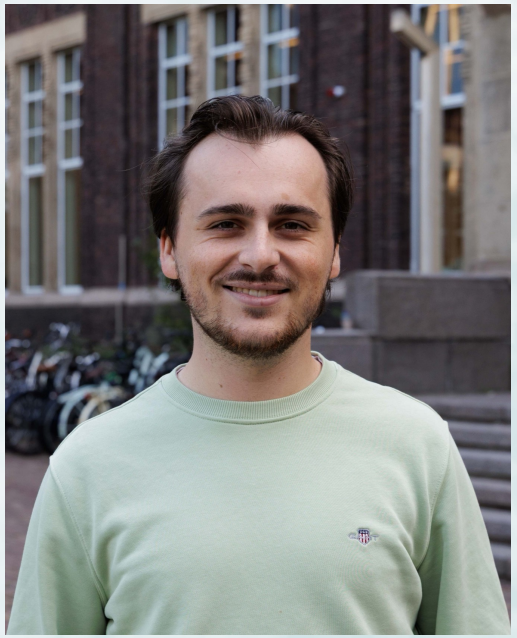
Arda Baysal
Artemi Kurski
Ruben Vons

TU Delft

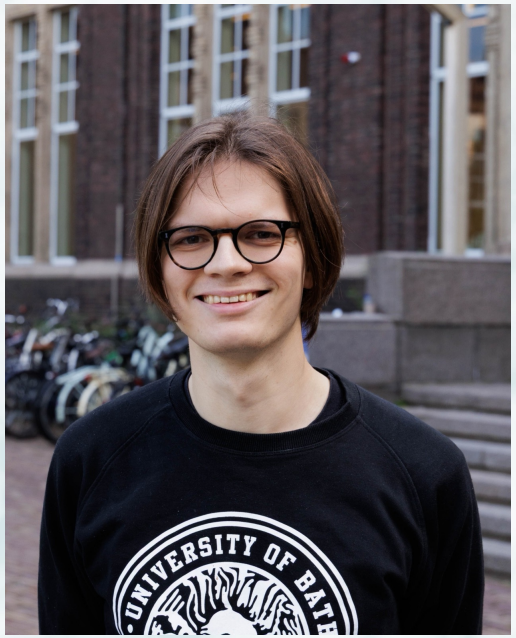


Project Team:

Arda Baysal



Ruben Vons



Artemi Kurski

Clients & Supervisors: Rijkswaterstaat, Deltares

- **Rijkswaterstaat**
  Government agency for infrastructure and water management.
- **Deltares**
  Independent research institue focused on water and subsurface research.
- **Shared Goals**
  The creation of a cross-border DEM for analysis and runoff simulation.

Context and Introduction: Why such an elevation model is useful?

- A Digital Elevation Model (DEM) represents height values over an area
  - **Digital Terrain Model (DTM):** only bare-earth elevation
  - Digital Surface Model (DSM): ground and all other objects on it including vegetation

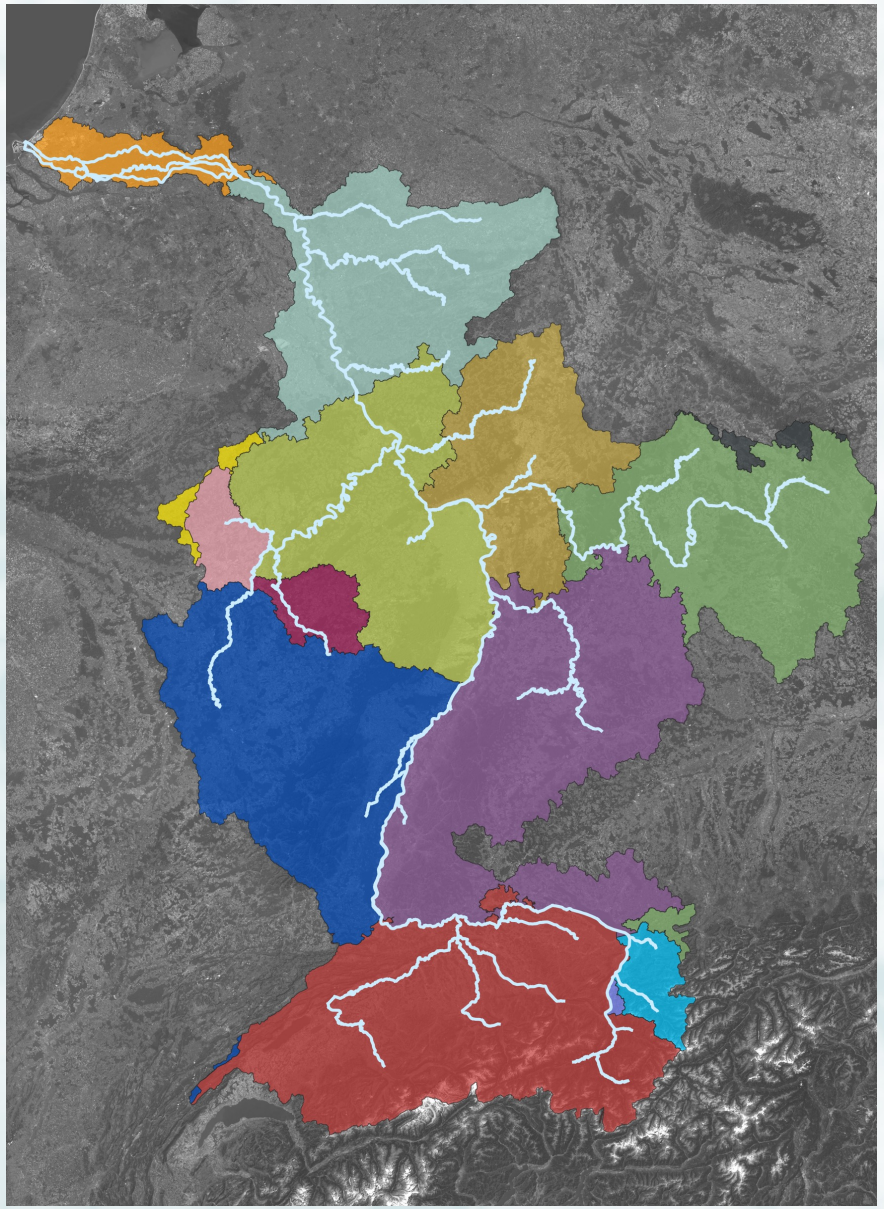
- Typical applications:
  - **Runoff and flood modelling**
  - Civil engineering
  - Defence
  - Ortho-imagery production

Problem Definition

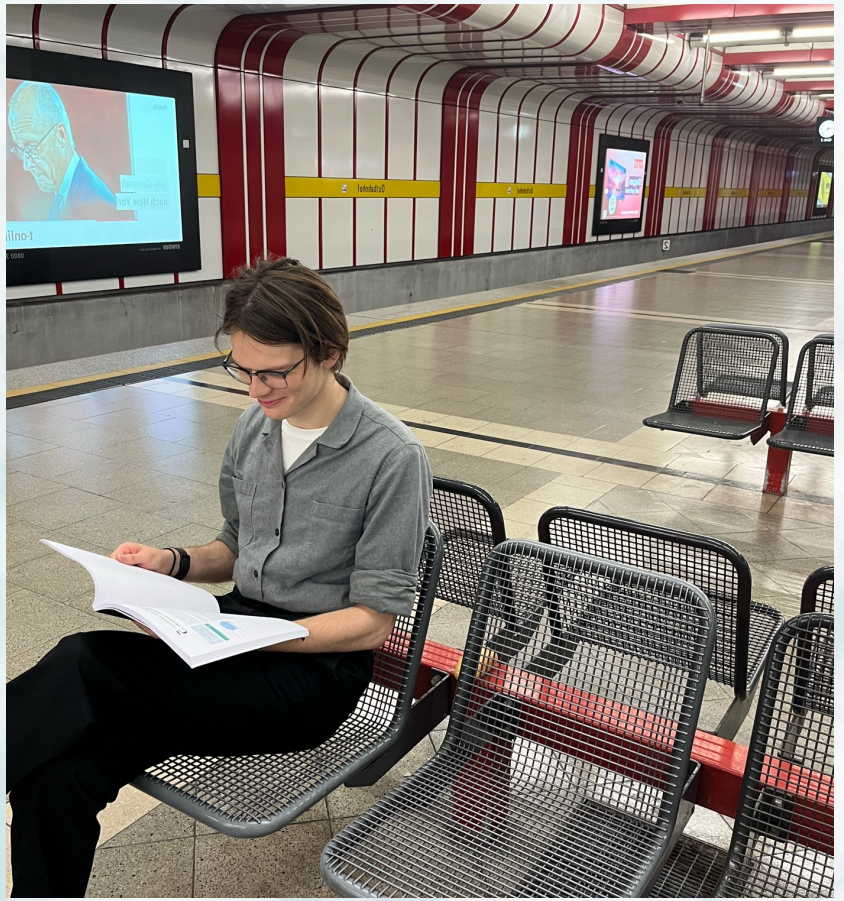
“The aim is to investigate how elevation data covering the Rhine catchment can be processed into a harmonised digital elevation model in an automated manner. The project considers the differences between the available source datasets and the processing steps required to bring them into a common representation.”

What is the Rhine Catchment Area?

- HydroBASINS definition is used
  - Includes all of upper and middle Rhine
  - Excludes the Meuse and parts of delta
- Countries covered:

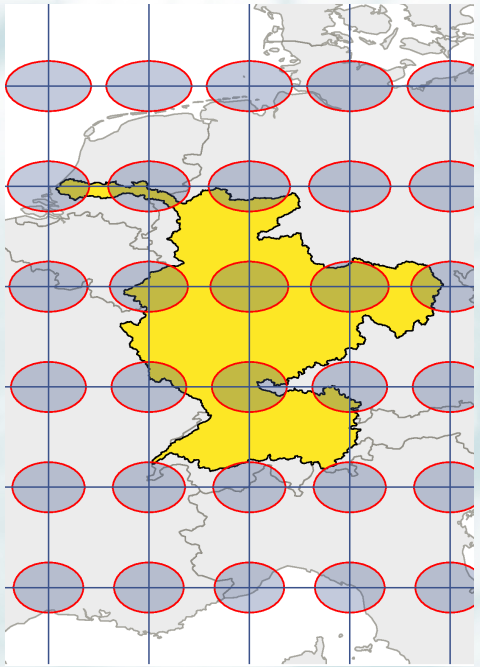


INSPIRE Specification on Elevation



- Bathymetry is separate from land-elevation
- Gridded (raster) data is required
- Tiles must have the same:
  - CRS,
  - encoding,
  - resolution and grid alignment.
- Requires ETRS89 with a matching datum.
- Recommends the European Vertical Reference System (EVRS)
- Relates horizontal resolution to vertical accuracy

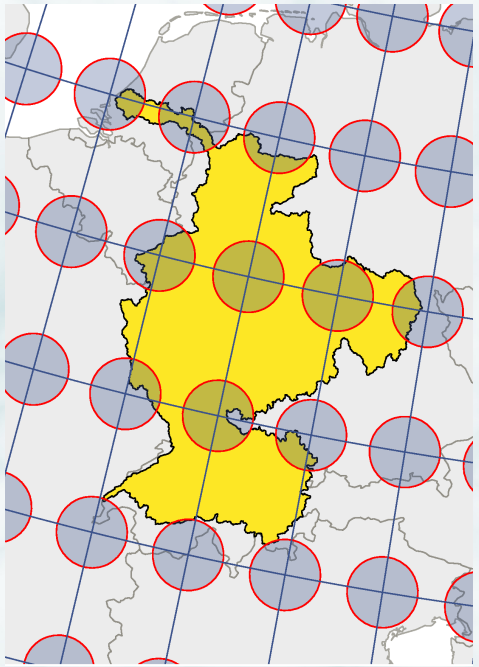
CRS Selection



ETRS89 (geodetic)

INSPIRE-compliant

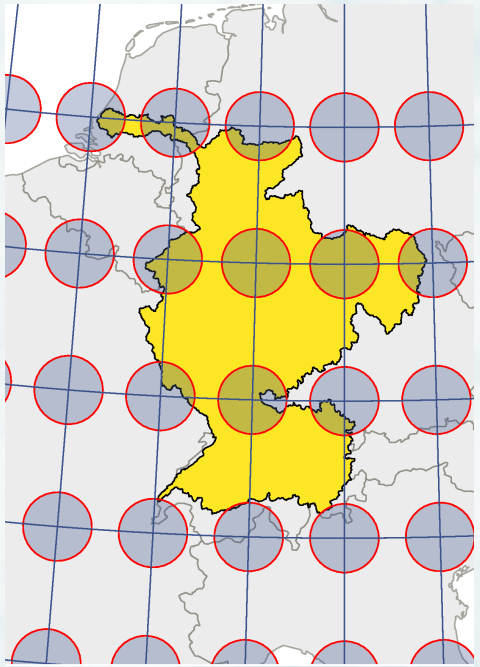
Does not preserve either shape or area



Equi7 Europe

INSPIRE-non-compliant

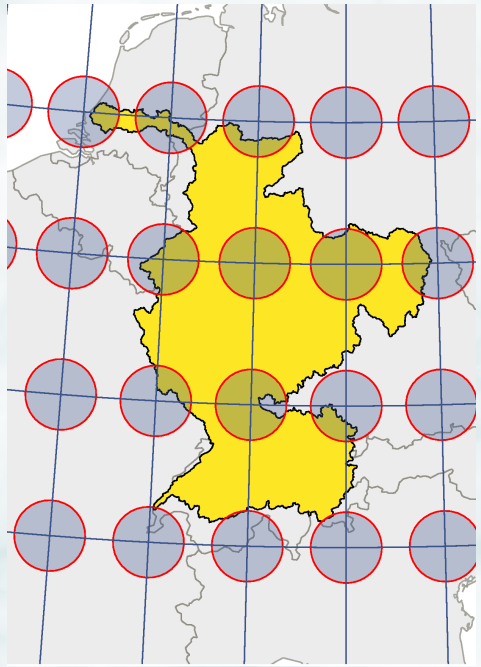
Compromises on preserving both shape and area



ETRS89-LCC

INSPIRE-compliant

Preserves shape, distorts area

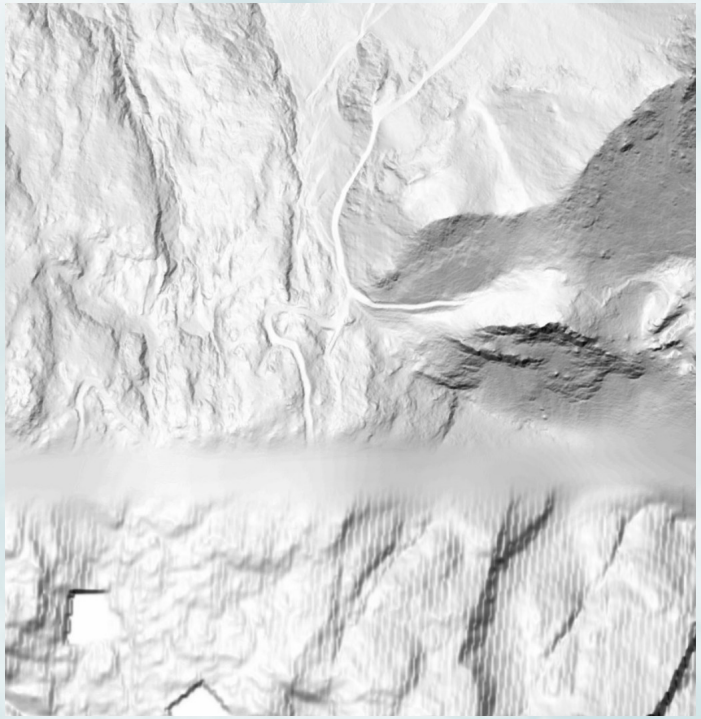


ETRS89-LAEA

INSPIRE-compliant

Preserves area, distorts shape

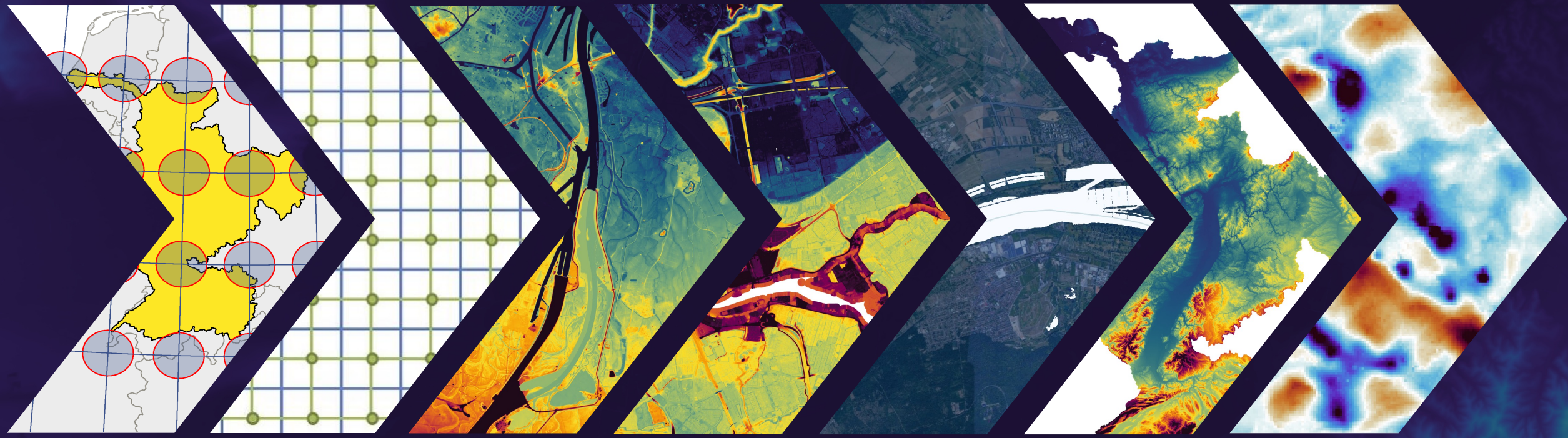
Review: Aggregate DEMs



Artefacts due to interpolation in Mapterhorn (Wipfli, 2026)

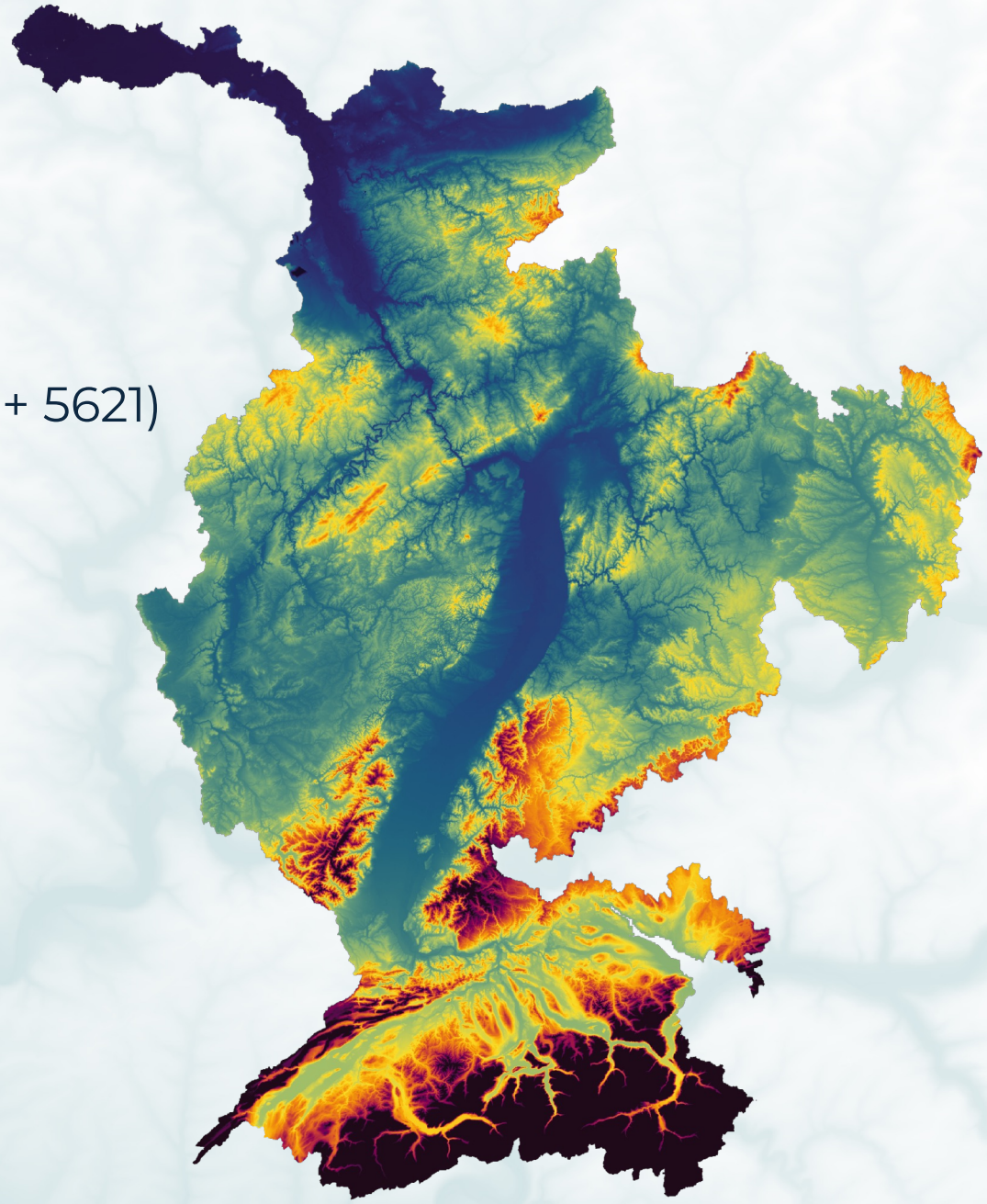
	Noardo et al., 2016	**Biagi & Carcano, 2013**	**Mapterhorn**
Region	Italy & France	Italy & Switzerland	Global
Data sources	National datasets	National datasets	National and global datasets
Resampling	Bilinear	Bicubic	Cubic spline
Border data harmonisation	Linear feathering	Vertical offset	Gaussian filter
Analysis	DEM difference	DEM difference	n/a
Validation	LiDAR data analysis	LiDAR data analysis	n/a

Data processing methodology



Results:
Catchment-Wide DEM

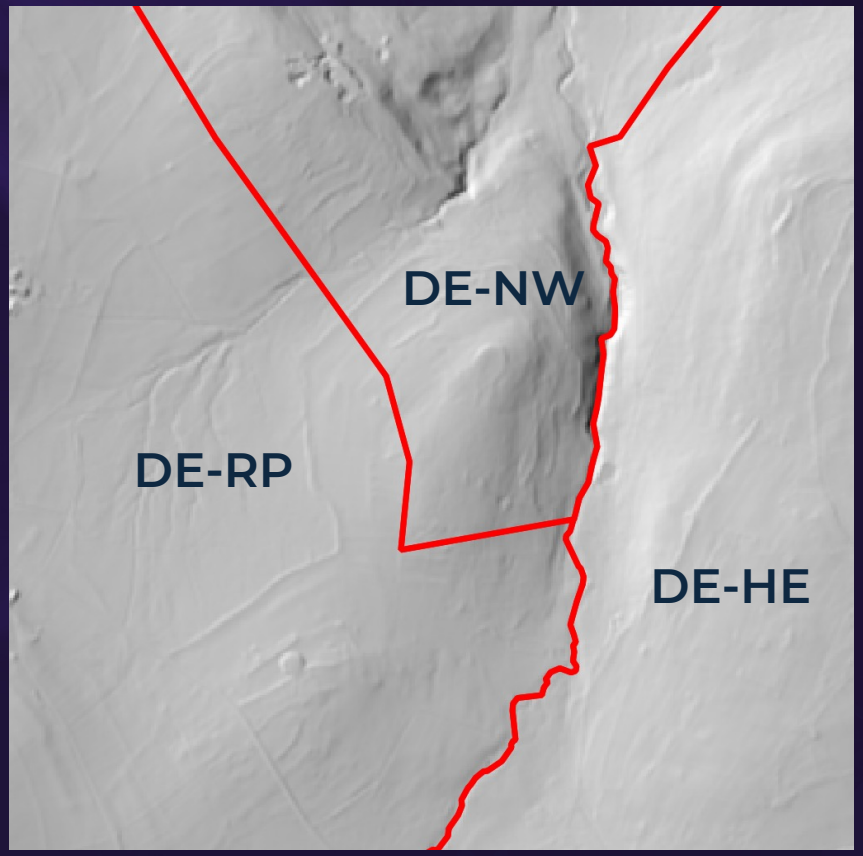
CRS: ETRS89-LAEA with EVRF-2007 (EPSG:3035 + 5621)
Resolution: 5m
Border data: not smoothed
Format: COG (cloud-optimised GeoTIFF)
File size: 17.5GB
Coverage: 159 284 km²
or 6 371 361 919 cells
Omission rate: 2.67%
Overlap rate: 2.60%, or 4140 km²
Elevation range: [-324.90m, 4272.68m]

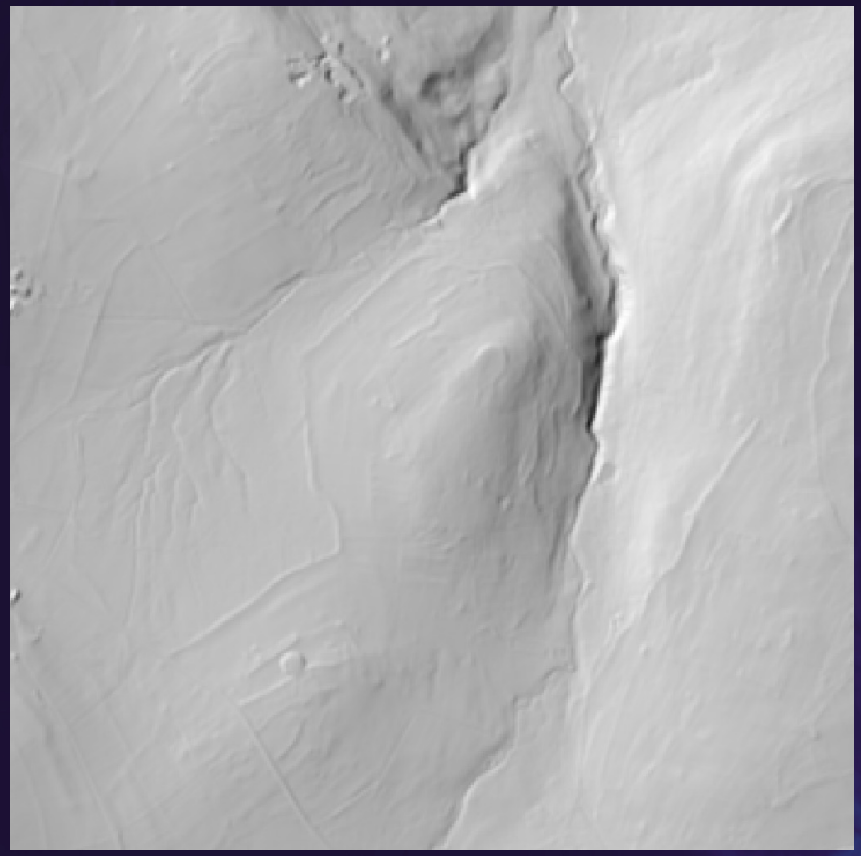


Processing Statistics

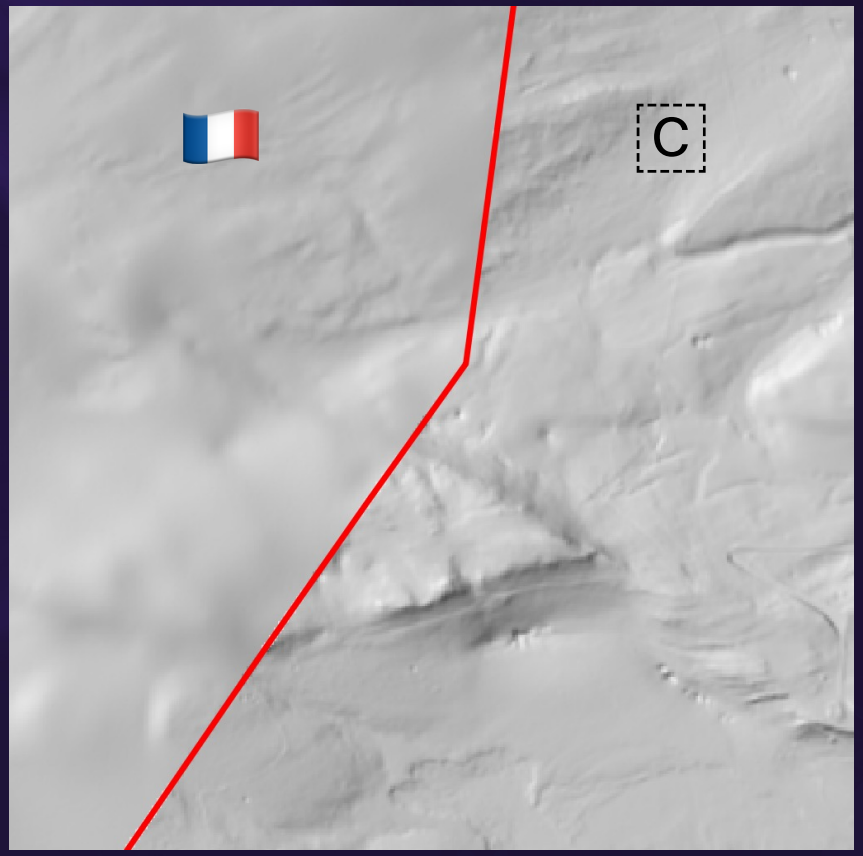
- **Hardware used:** 3D Geoinformation server
  - 128 threads at 2.9GHz clock
  - 512 GB RAM
- **Implementation:**
  - maximum 64 threads used
  - 4 parallel processes for resampling each country's DEM
  - sequential processing otherwise
- **Time to completion:**
  - 2:29:22 to aggregate DEM
  - 14:44:39 for analysis

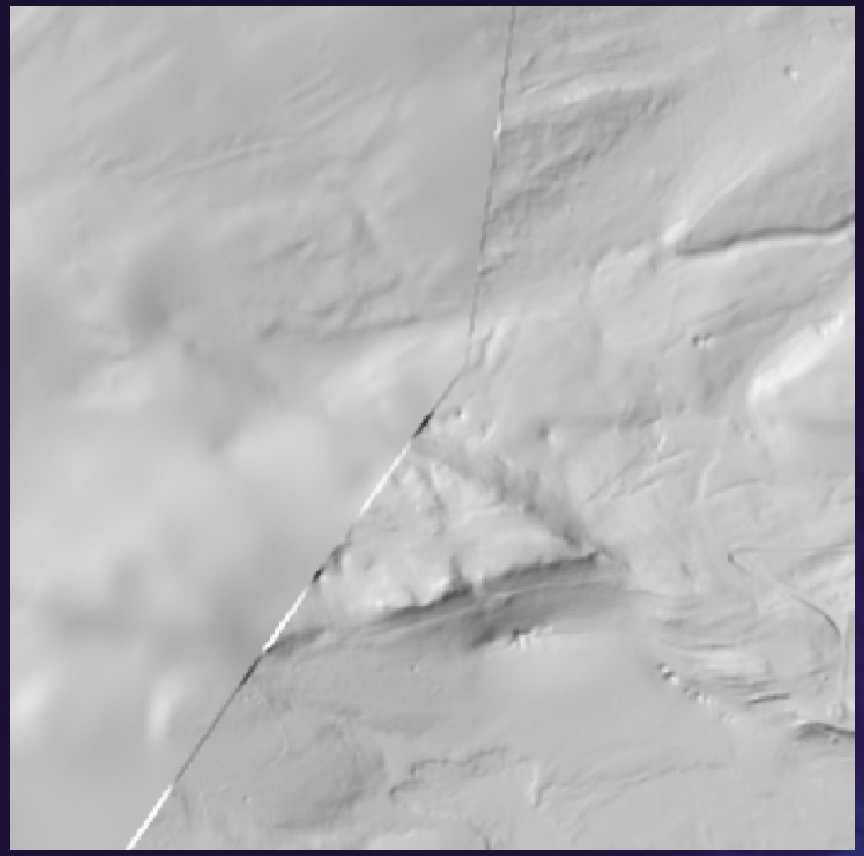
DTM seamlessness: German states





DTM seamlessness: France/Switzerland

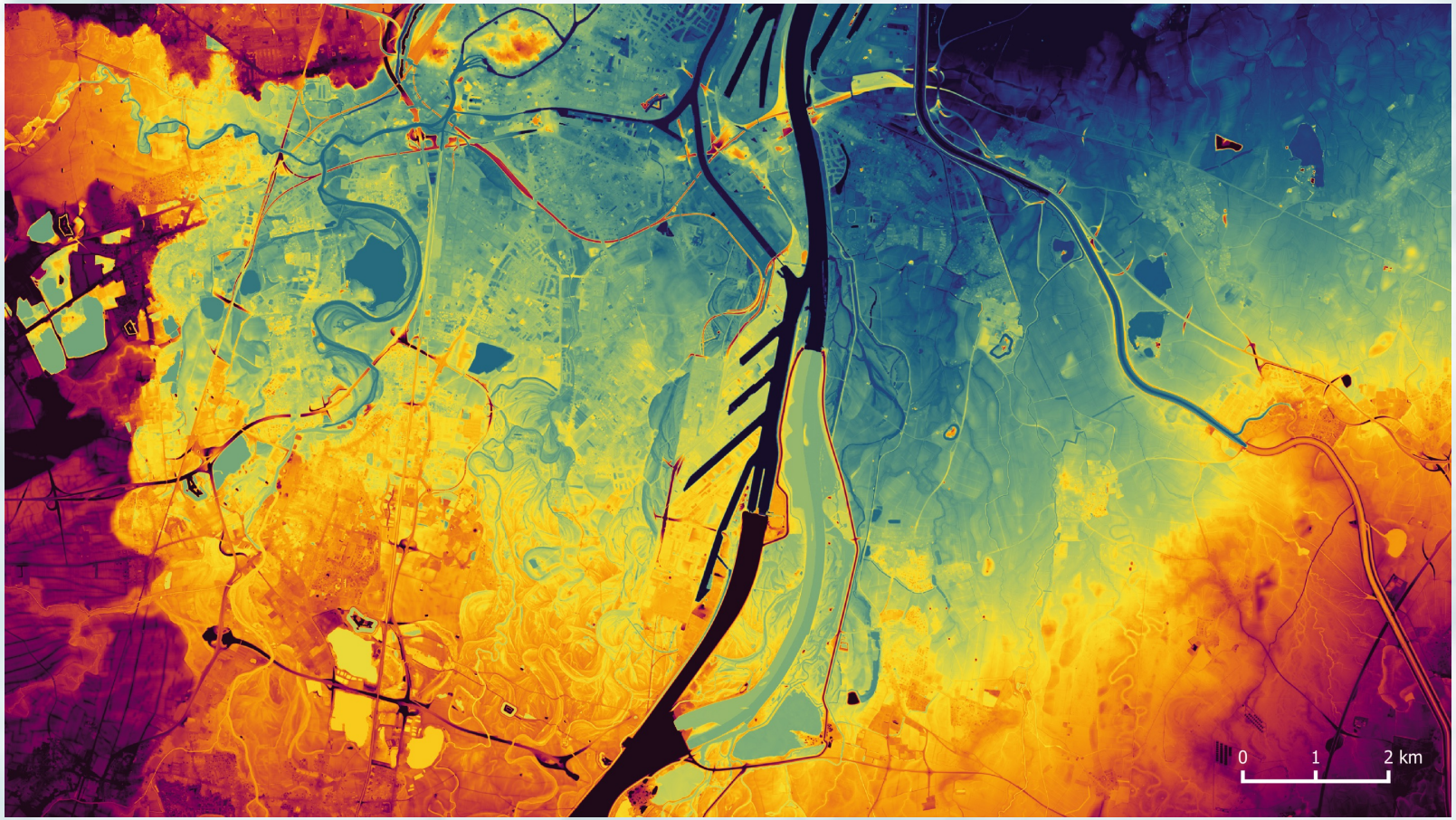




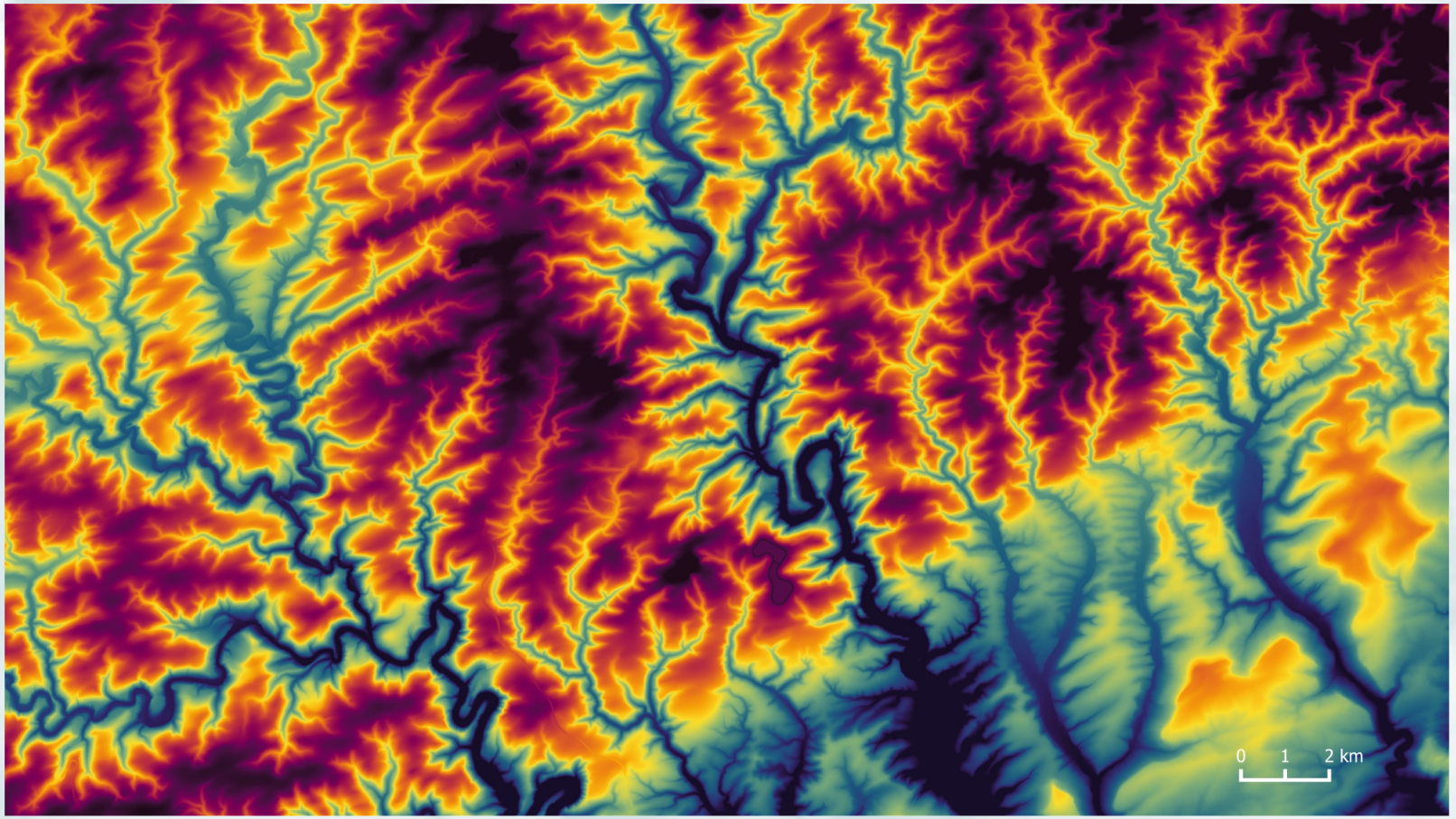
Source data specifications

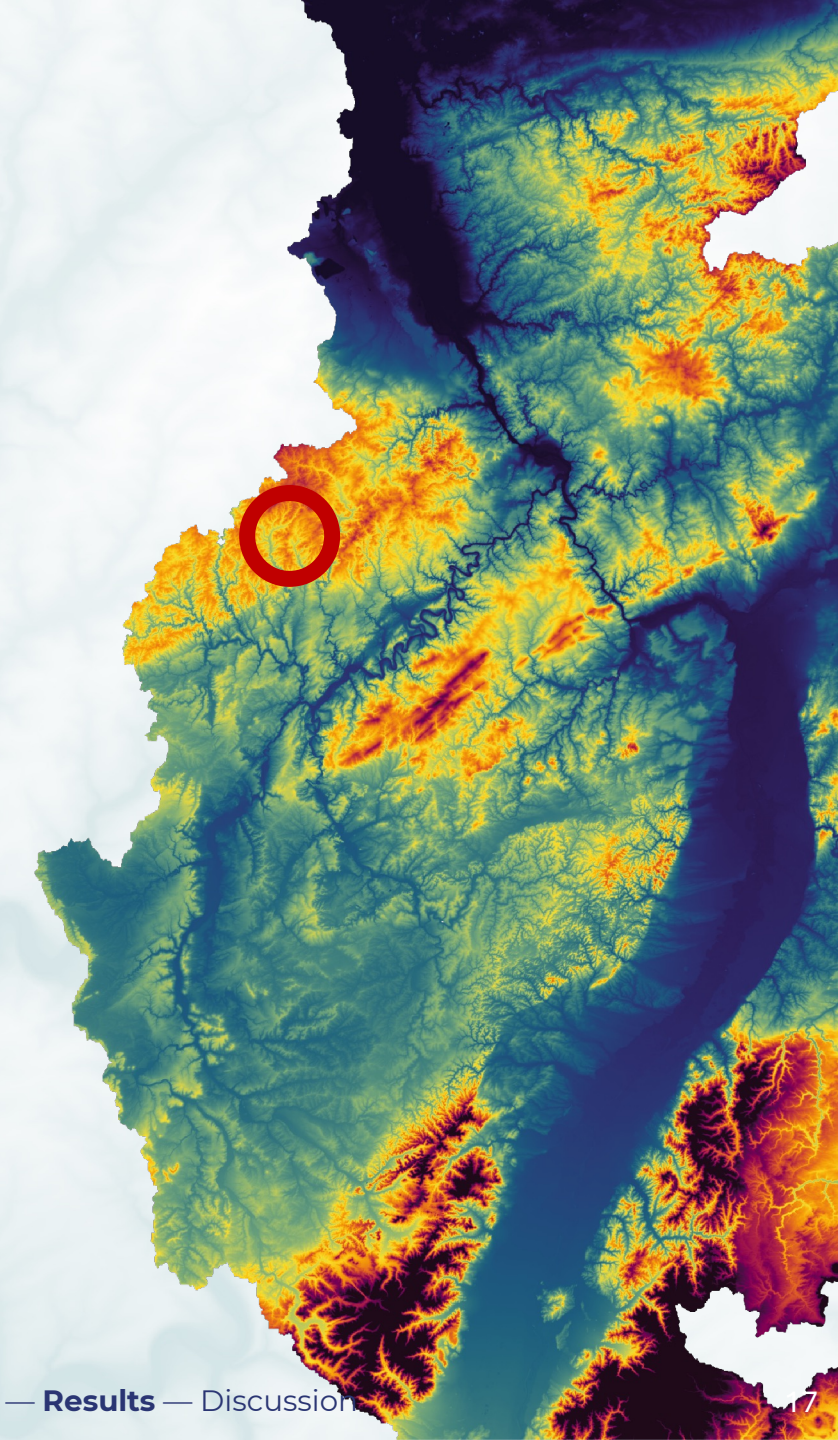
Region	Dataset / provider	Source format	Survey method	Native grid (m)	CRS (EPSG)	Vertical datum
Netherlands	AHN / Rijkswaterstaat	GeoTIFF	ALS	0.5	28992	NAP
Bavaria (DE)	DGM5 / LDBV	XYZ Tiles	ALS	5	25832	DHHN2016 / NHN
Baden-Württemberg (DE)	DGM1 / LGL	XYZ Tiles	ALS	1	25832	DHHN2016 / NHN
Hessen (DE)	ATKIS DGM1 / HVBG	GeoTIFF	ALS	1	25832	DHHN2016 / NHN
North Rhine-Westphalia (DE)	DGM1 / Geobasis NRW	GeoTIFF	ALS	1	25832	DHHN2016 / NHN
Rhineland-Palatinate (DE)	DGM1 / LVermGeo	GeoTIFF	ALS	1	25832	DHHN2016 / NHN
Saarland (DE)	DGM1 / LVGL	GeoTIFF	ALS	1	25832	DHHN2016 / NHN
France	RGE ALTI 5 m / IGN	ASCII Grid	Mixed[1]	5	2154	IGN69
Luxembourg	LiDAR 2024 MNT / ACT	GeoTIFF	ALS	0.5	2169	n/a
Liechtenstein	swissALTI3D / swisstopo	GeoTIFF	Mixed[2]	2	2056	LN02
Switzerland	swissALTI3D / swisstopo	GeoTIFF	Mixed[2]	2	2056	LN02

Close-up examples

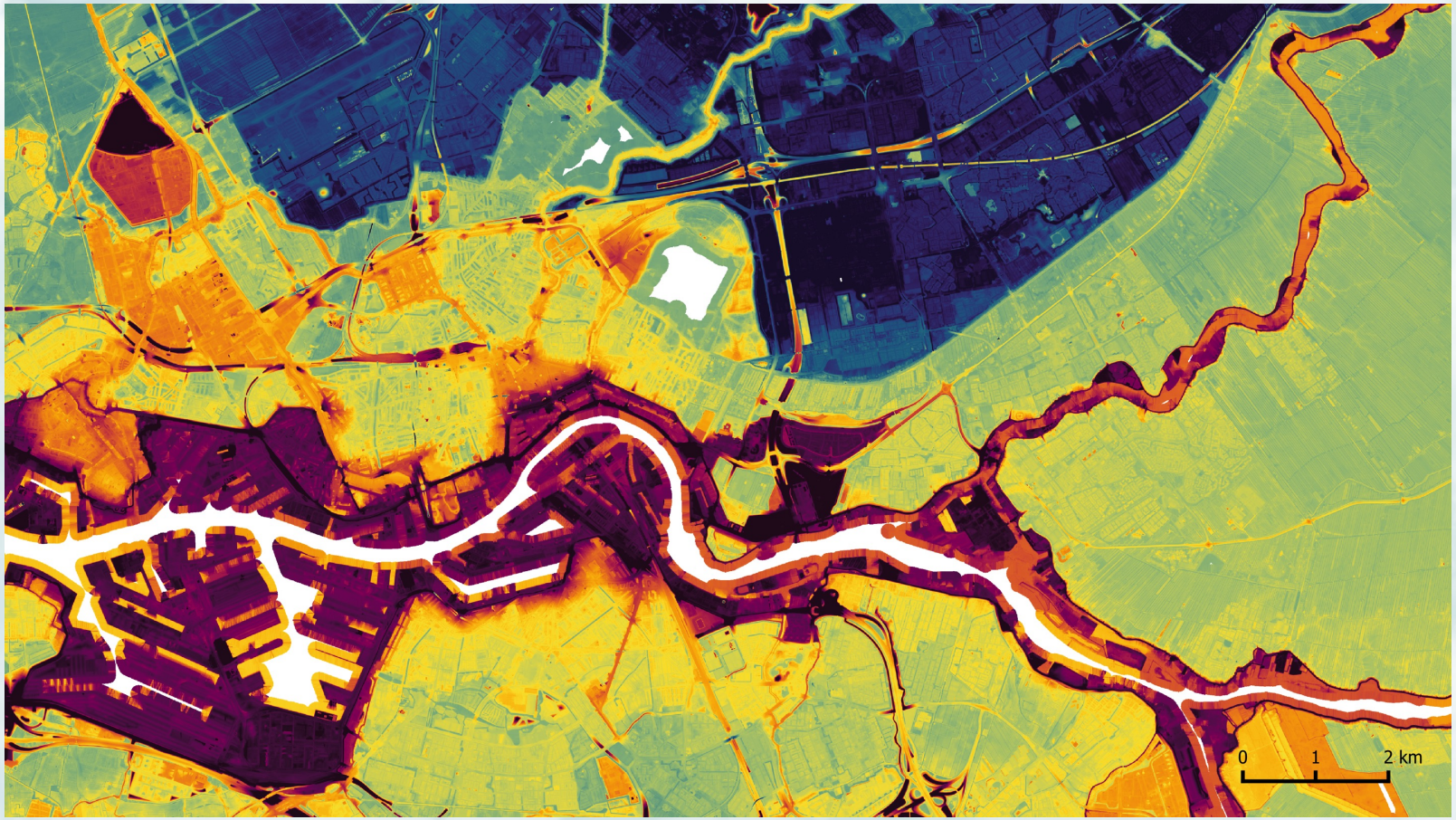


Close-up examples



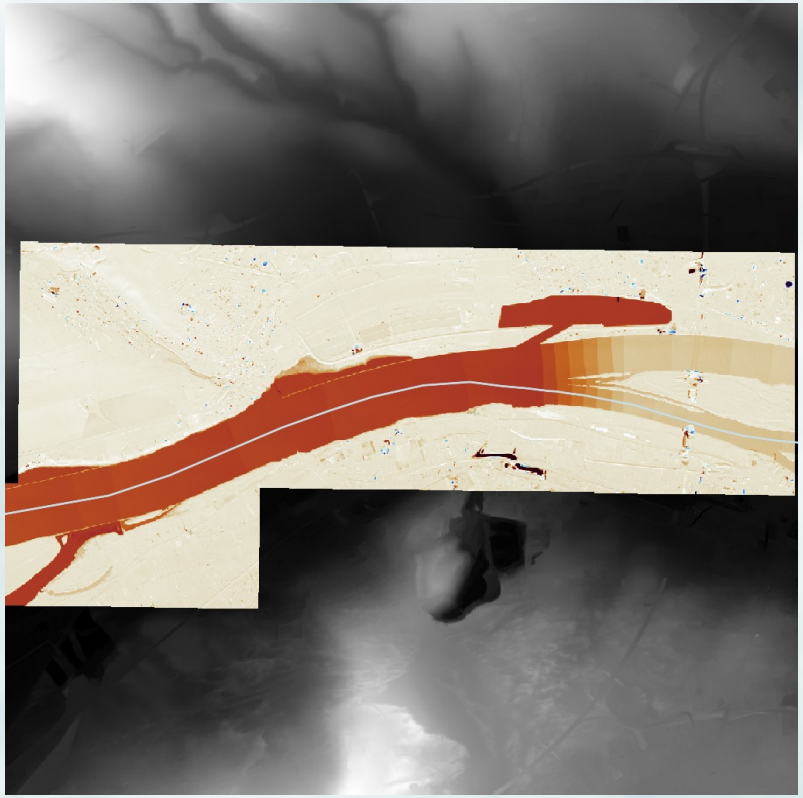


Close-up examples



Creating a water mask

Wiesbaden,Hessen, Germany



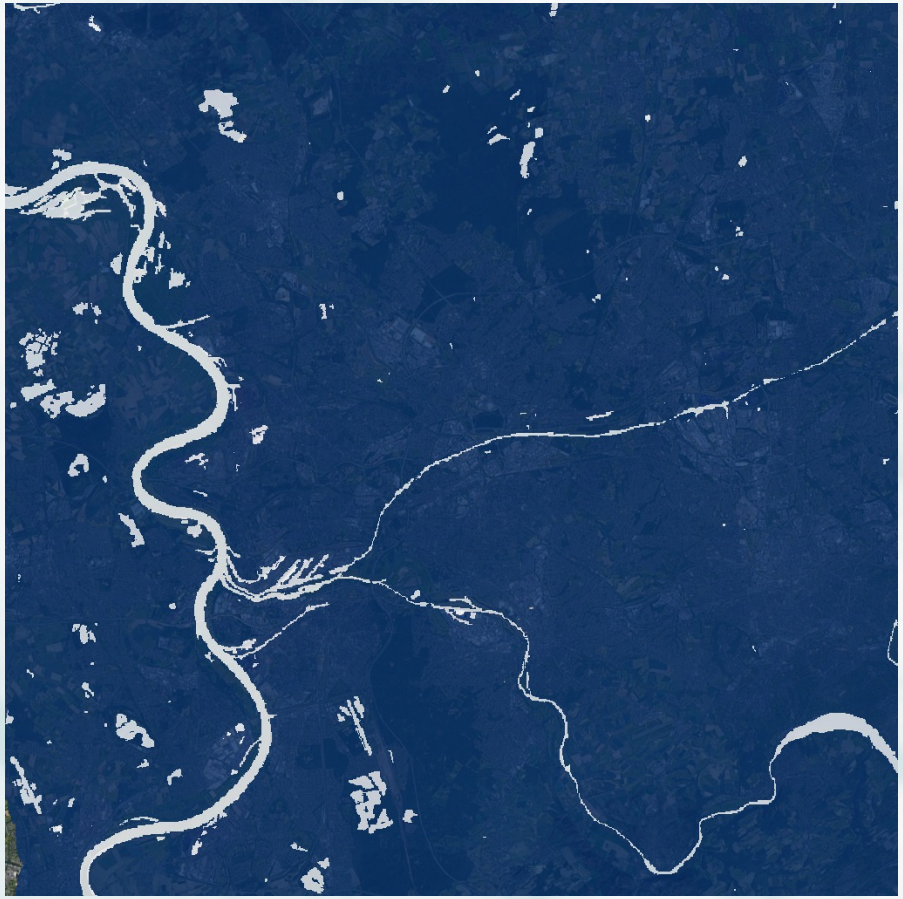
Difference raster

Coarse water mask

Finer water mask

Creating a water mask

Rhine catchment



Cologne area

Analyzing differences in DTMs

Border pair	mean	median	rmse	le90	min	max
France/Luxembourg	0.658	0.584	3.129	3.827	-23.190	39.686
France/Switzerland	0.095	0.101	0.727	0.405	-24.450	15.926
Germany-BW/France	-0.479	-0.458	0.873	0.779	-26.461	46.219
Germany-BW/Germany-Bavaria	-0.053	0.039	5.361	0.114	-961.151	29.982
Germany-BW/Switzerland	0.047	0.048	0.406	0.124	-21.185	34.454
Germany-Hessen/Germany-Bavaria	0.003	0.002	0.178	0.098	-18.354	19.075
Germany-Hessen/Germany-BW	0.011	0.009	0.112	0.064	-10.628	14.753
Germany-NRW/Germany-Hessen	0.010	0.011	0.184	0.075	-9.371	22.737
Germany-NRW/Germany-RLP	-0.028	-0.023	0.195	0.196	-22.004	31.373
Germany-RLP/France	-0.559	-0.559	0.616	0.715	-30.898	34.020
Germany-RLP/Germany-BW	-0.009	-0.013	0.210	0.073	-19.532	16.480
Germany-RLP/Germany-Hessen	0.013	0.045	1.297	0.186	-61.556	30.440
Germany-RLP/Germany-Saarland	0.008	0.011	0.374	0.148	-38.308	26.917
Germany-RLP/Luxembourg	0.032	0.027	0.353	0.228	-25.563	17.085
Germany-Saarland/France	-0.514	-0.563	1.317	1.161	-22.604	32.059
Germany-Saarland/Luxembourg	-0.016	-0.003	0.426	0.166	-10.565	10.442
Netherlands/Germany-NRW	-0.065	-0.063	0.244	0.134	-11.543	13.759

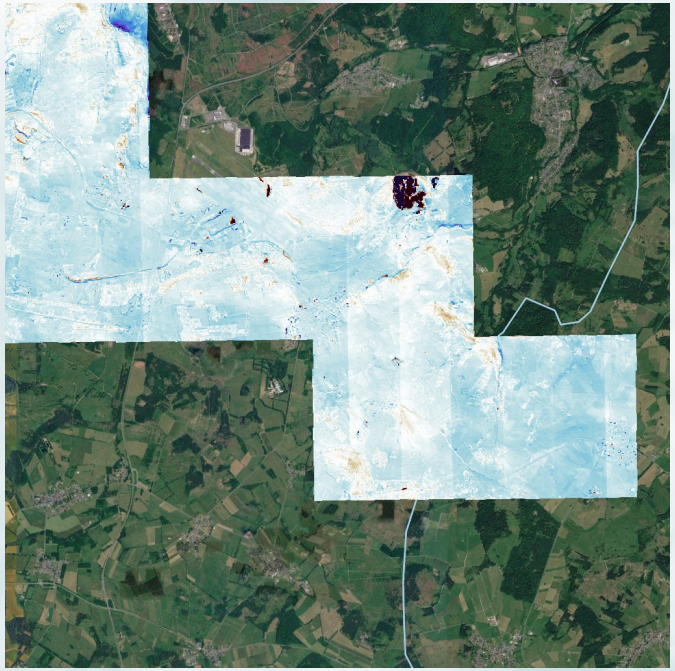
Analyzing differences in DTMs

Category / subcategory	Border pairs	Percentage of pairs	Occurance Rank	Severity rank
Interpolation	17	94%	1	5
Interpolation diffences	17	94%		
Ground filtering	15	83%	2	2
Slopes	13	72%		
Overpass	5	28%		
Temporal	14	78%	3	3
Construction	11	61%		
Quarry	10	56%		
Storage	5	28%		
Snow	1	6%		
Watermask	13	72%	4	4
River mask failure	13	72%		
Water handling	5	28%		
Border	7	39%	5	1
Border artifacts	5	28%		
Extrapolation	4	22%		
Tile artifacts	7	39%	5	6
Tile differences	4	22%		
Acquisition striping	4	22%		

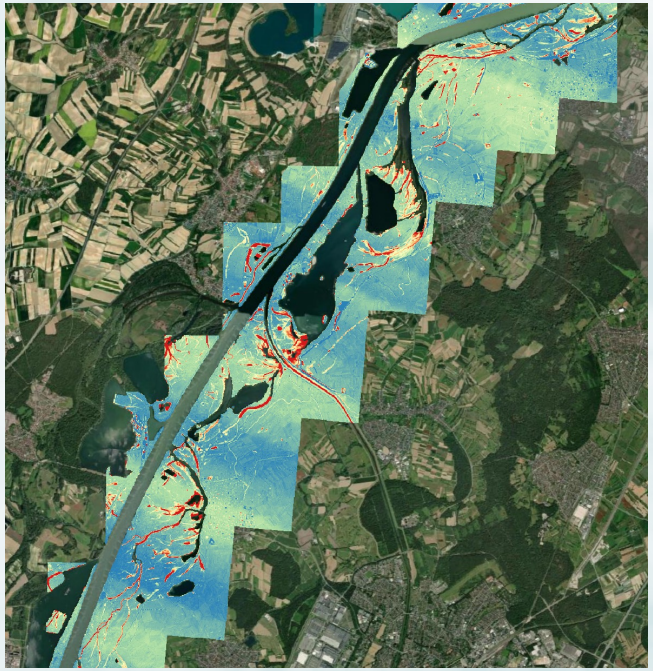
Analyzing differences in DTMs

- Tile Artefacts - Flight patterns

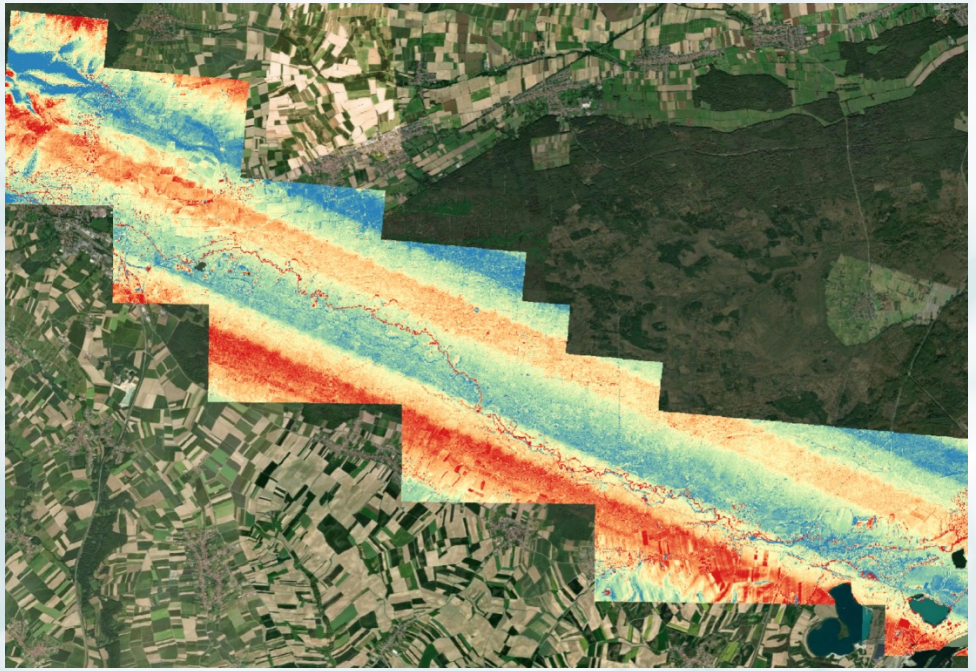
Elevation difference rasters



NRW - RLP



BW - France

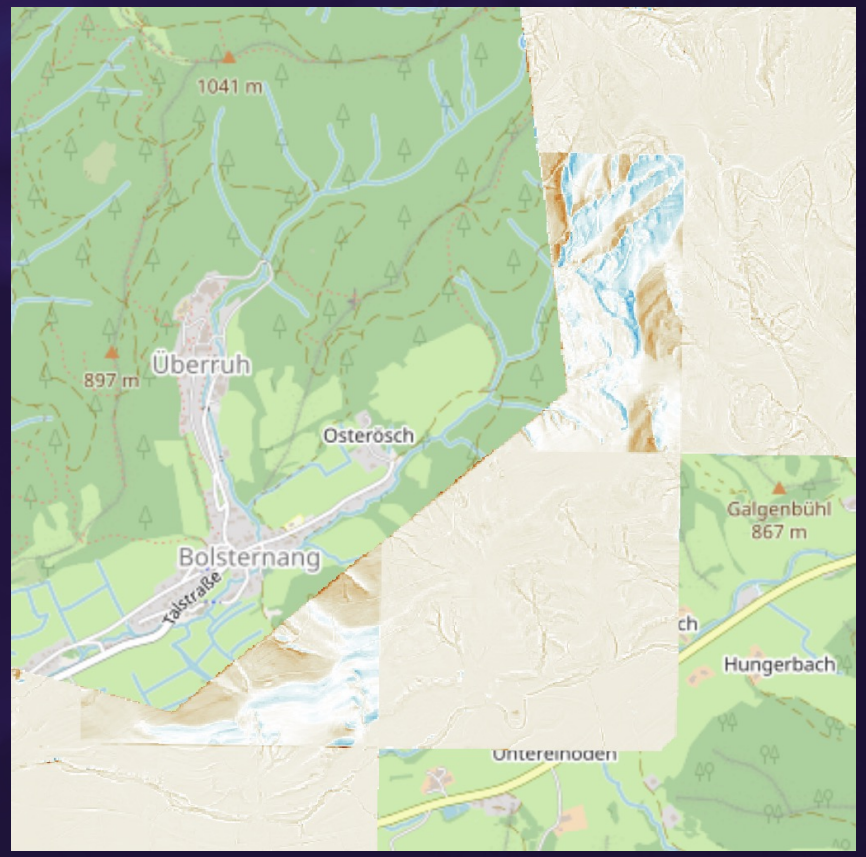


RLP - France

Analyzing differences in DTMs

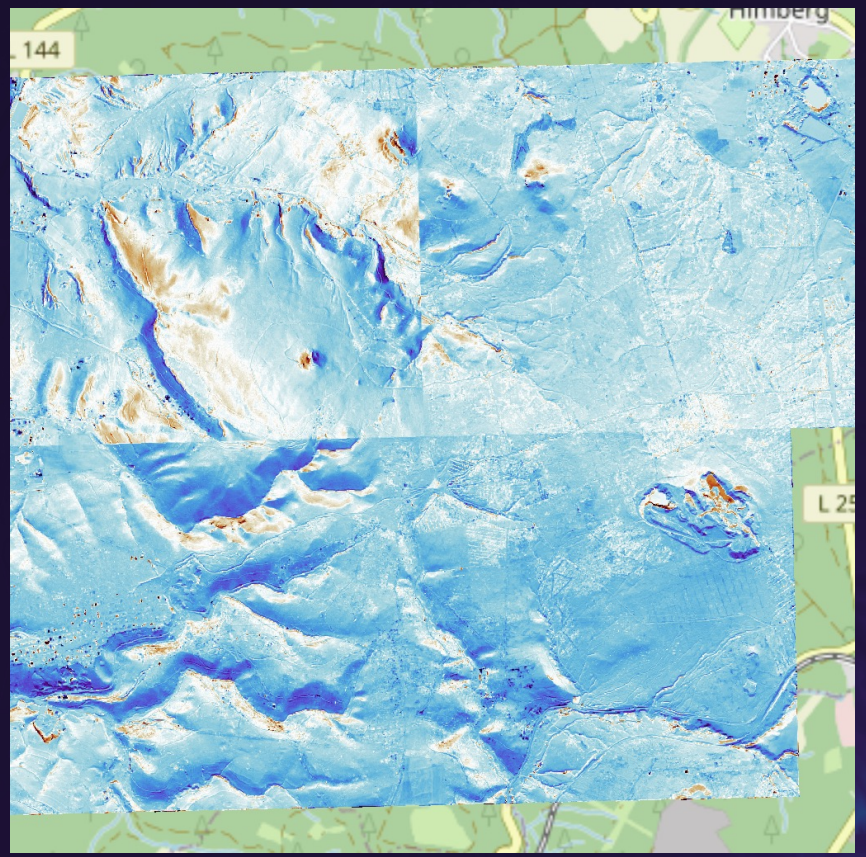
- Tile Artefacts – Tile seam patterns

Uberruh – Baden-Wurttemberg – Germany



Elevation difference raster

Leyberg – Nordrhein-Westfalen – Germany

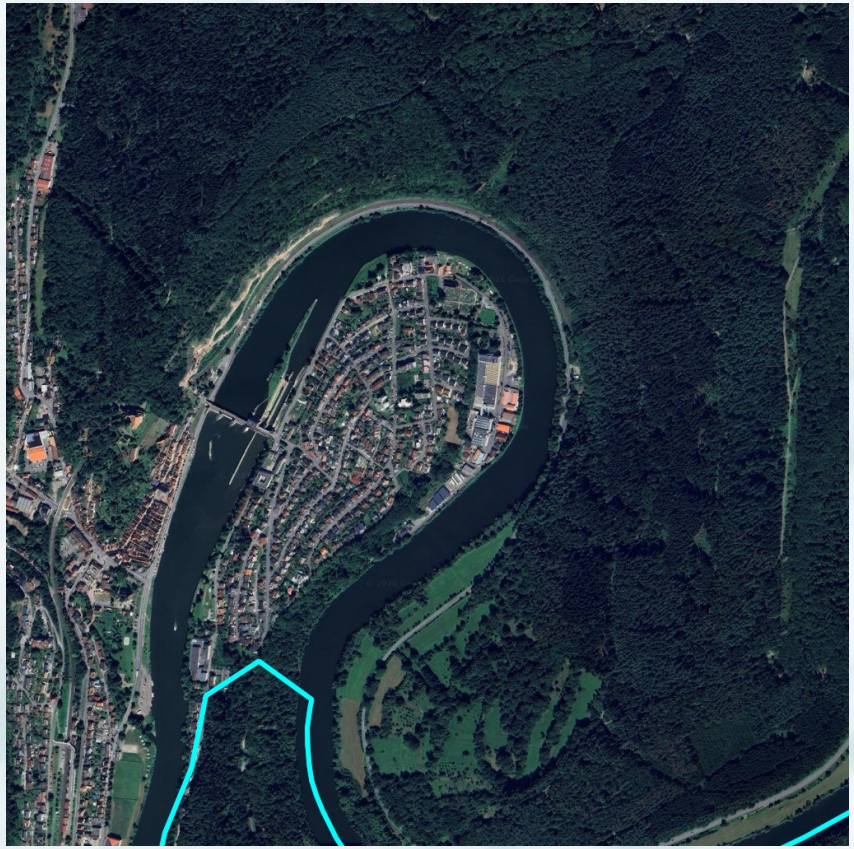


Elevation difference raster

Analyzing differences in DTMs

- Interpolation artefacts

Elevation difference raster



Satellite Imagery

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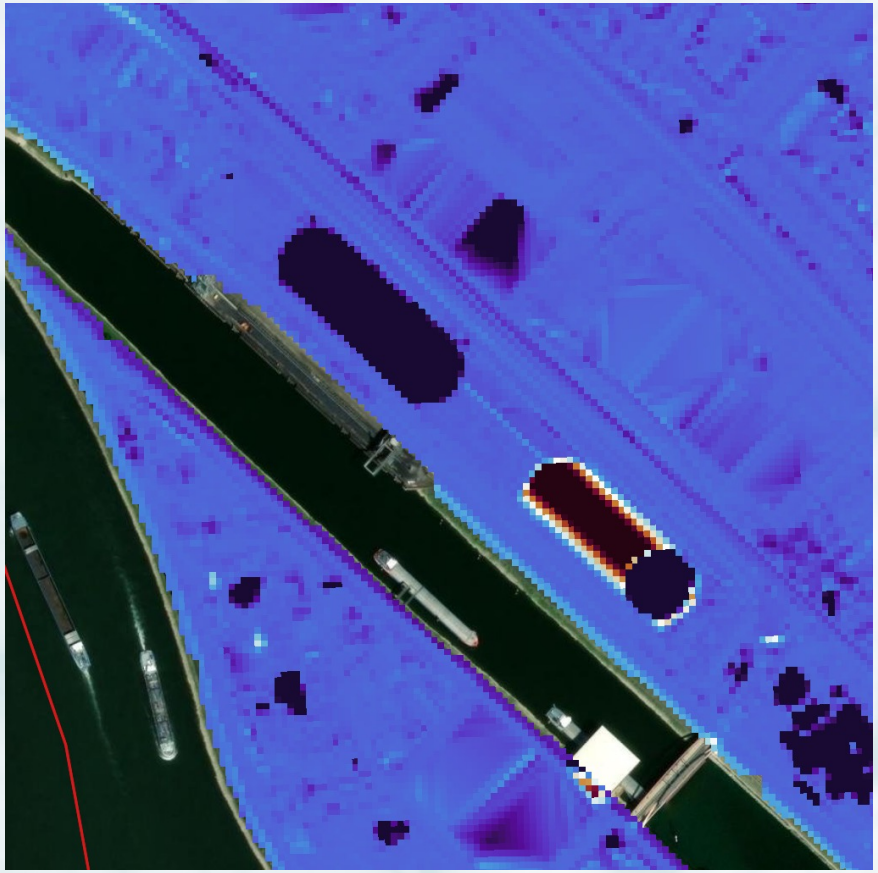
Analyzing differences in DTMs

- Temporal artefacts – Earthworks and storages

Cologne – Baden-Wurttemberg – Germany

Satellite Imagery

©2026 Airbus, Landsat / Copernicus, Maxar Technologies



Elevation Difference

Analyzing differences in DTMs

- Temporal artefacts – Quarries

Herblingen – Switzerland

Satellite Imagery

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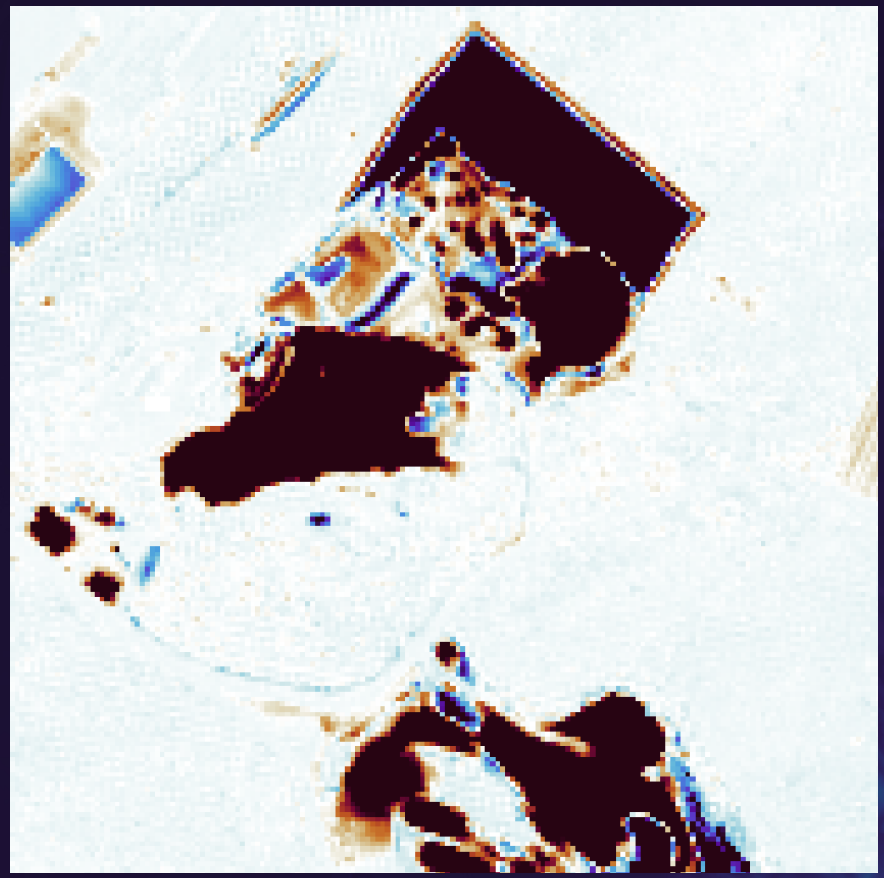
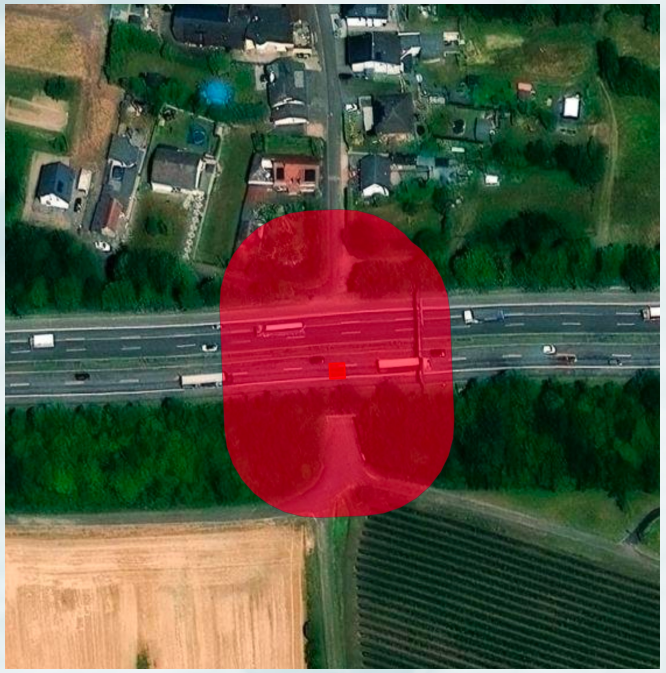


Elevation Difference

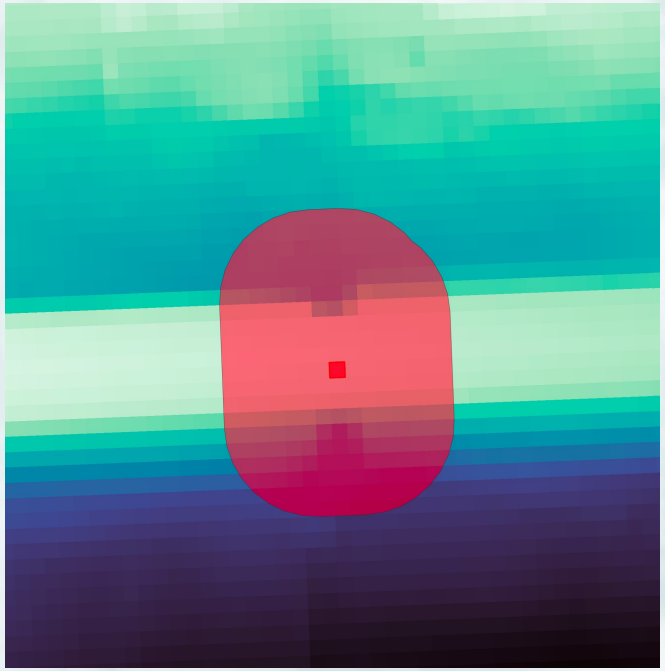
Analyzing differences in DTMs

- Ground filtering artefacts - Overpasses

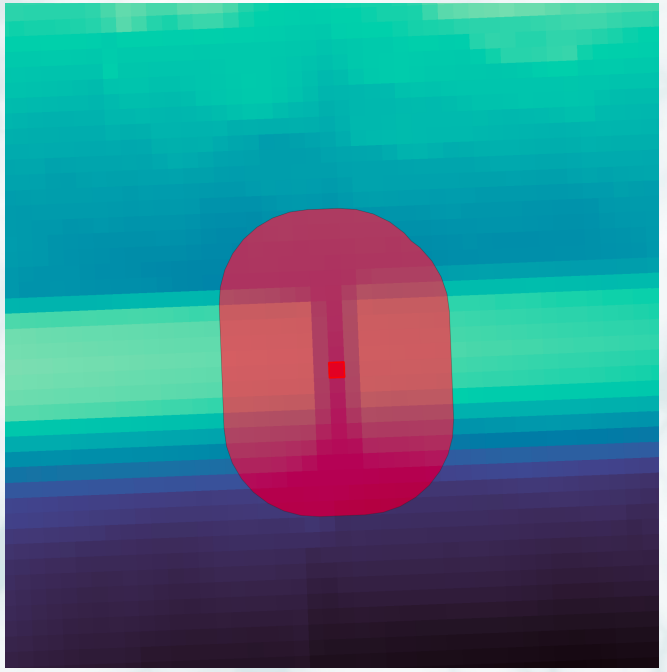
Beller – Rheinland-Pfalz – Germany



Satellite Imagery



Nordrhein Westfalen



Rheinland Pfalz

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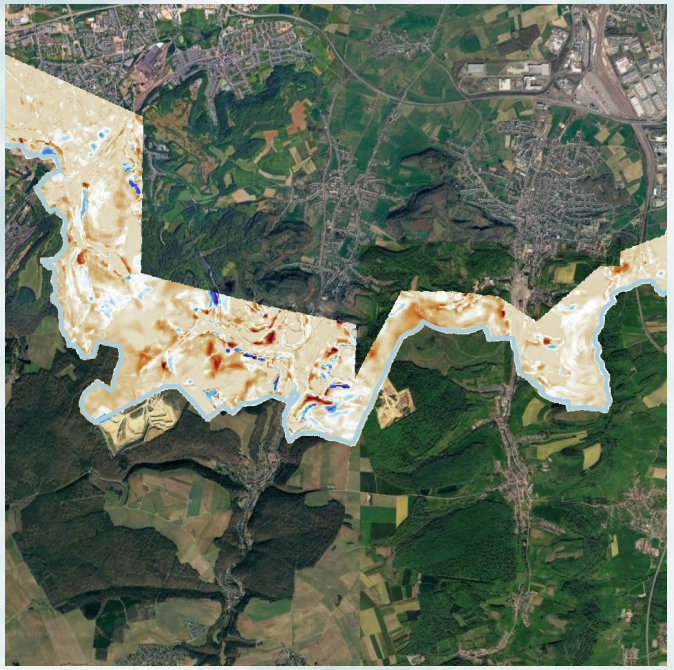
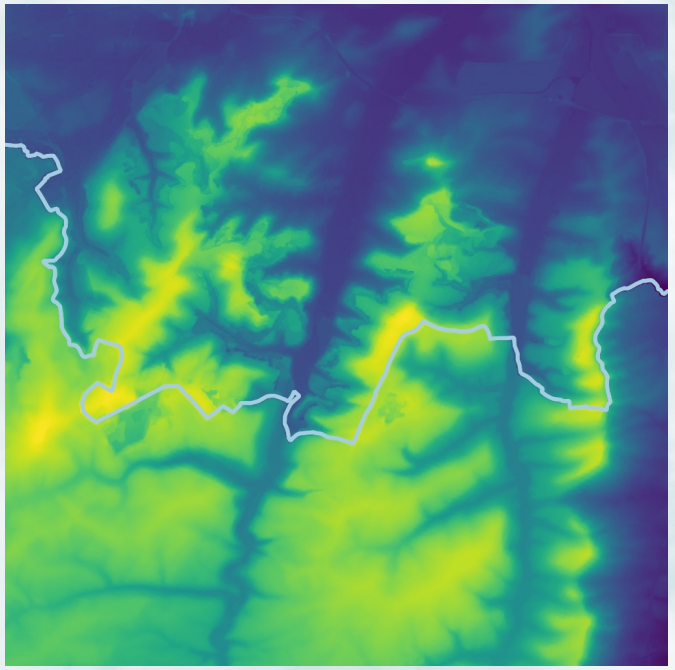
Analyzing differences in DTMs

- Ground filtering artefacts – Valleys

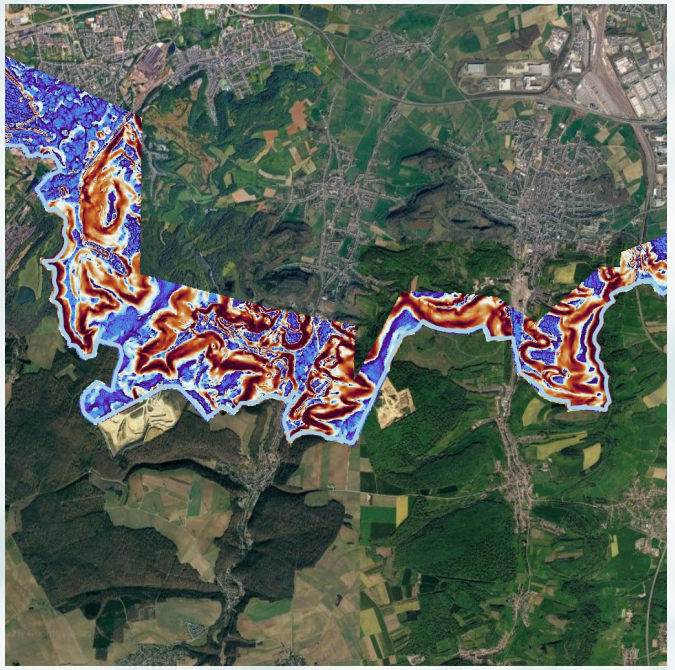
Rumelange – Luxembourg



Elevation difference raster



DTM

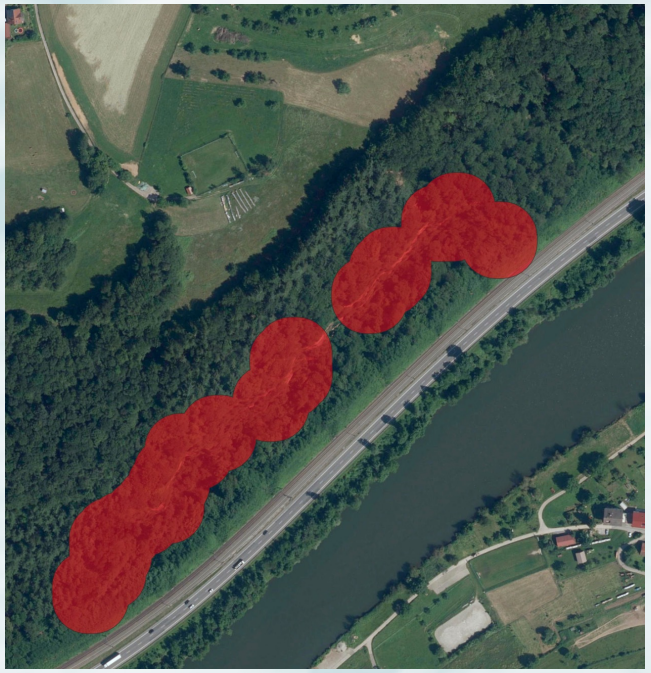


Random Errors

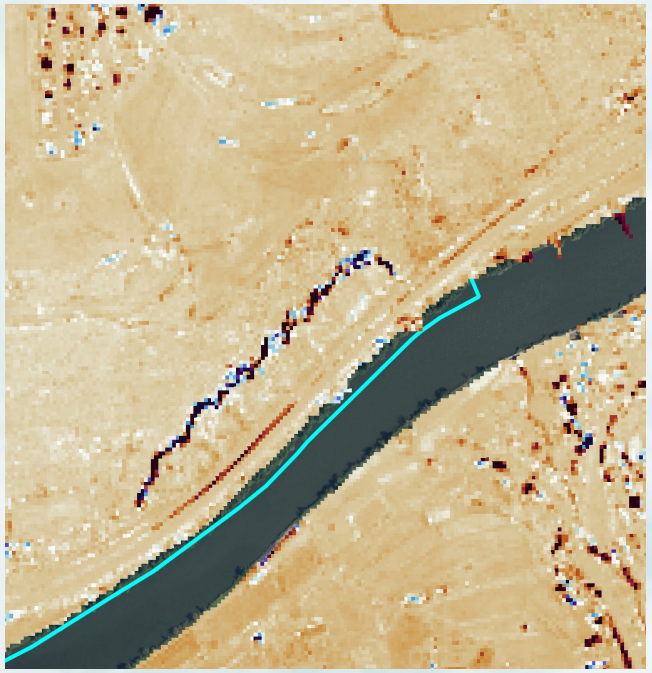
Analyzing differences in DTMs

- Ground filtering artefacts – Hilly surfaces and cliffs

Pleutersbach – Baden-Wurttemberg – Germany



Satellite Imagery



Elevation difference raster

Google Street View

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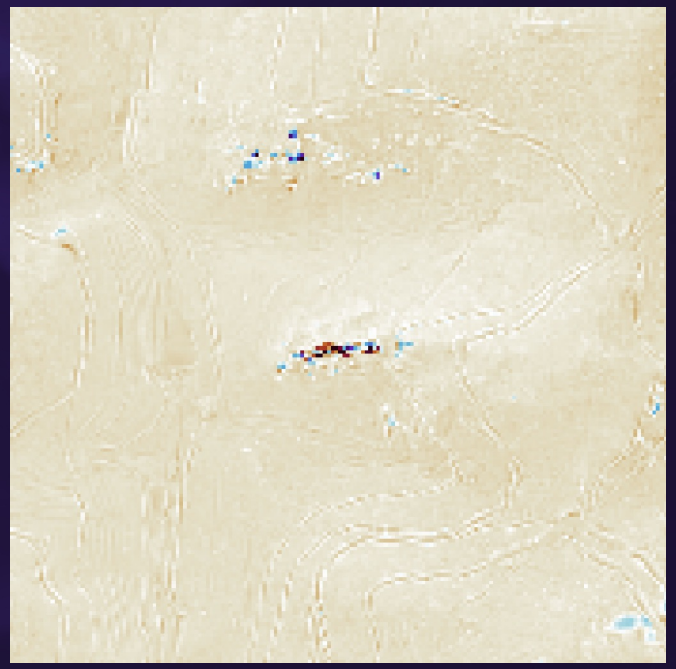
Analyzing differences in DTMs

- Ground filtering artefacts – Castle ruins

Klein-Arsberg, Ober Steinbach, France



Elevation difference

Castle ruins

Analyzing differences in DTMs

- Extrapolation artefacts

Hambach – Rheinland-Pfalz – Germany

Satellite Imagery

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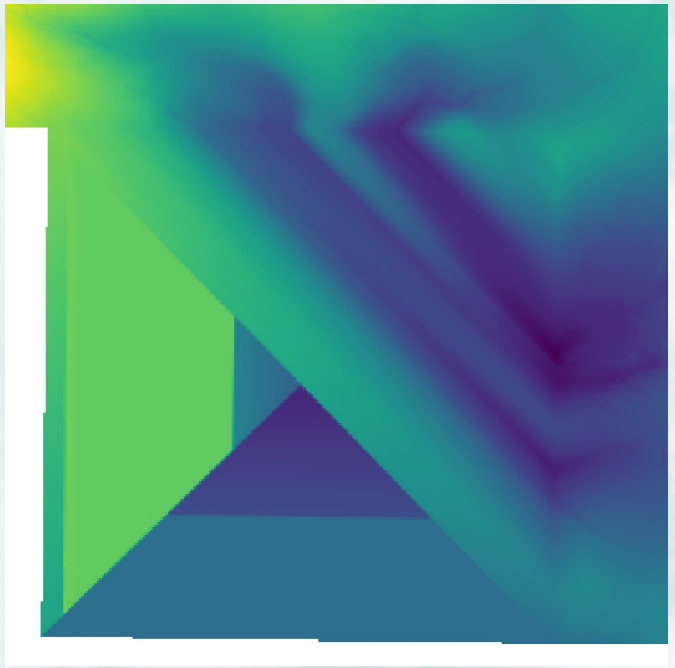
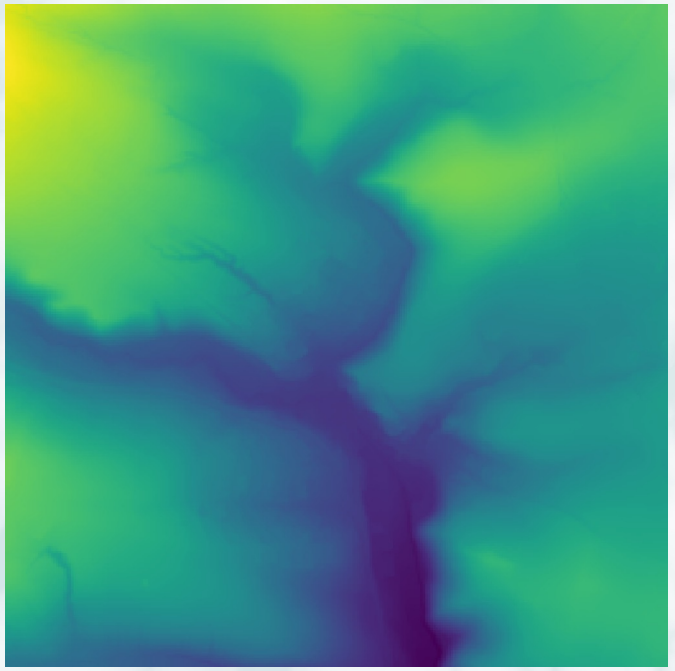


Hessen DTM



Rheinland Pfalz DTM

Discussion & recommendations

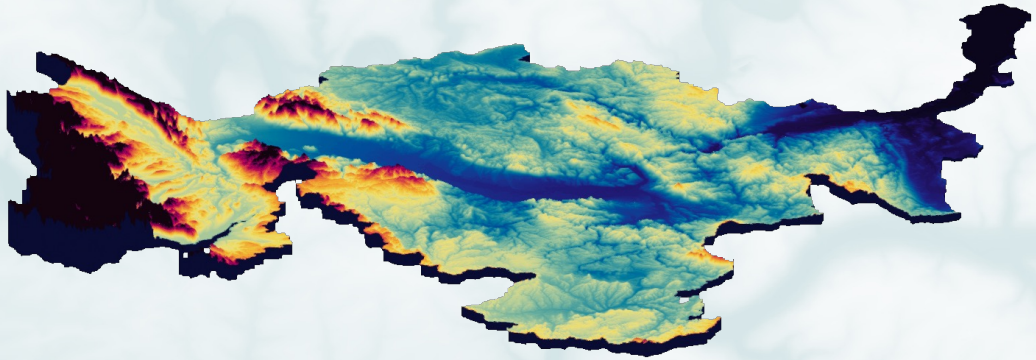
- A (mostly) seamless Rhine Catchment DTM has been successfully created
- Most of the catchment area covered
- A viable alternative to GDEMs / Mapterhorn
- A lot of differences arise from different ground filtering
- Using the newest & most accurate DTM product can improve results

Limitations:

- Sensitive to quality of input data
- Not fully seamless

Recommendations:

- Using a common European CRS
- Standardising the processing



Thank you for your attention. Questions?

Arda Baysal – Artemi Kurski – Ruben Vons
TU Delft MSc Geomatics